

# The Ising gap gives a conditional local parent Hamiltonian

The square-root Gibbs transform of C124 is a positive, three-site local, frustration-free Hermitian operator with unique ground state  $\sqrt{\pi_{N,b}}$  and exact gap  $a[1 - \tanh(2b)]$ . Multiplication by a supplied action constant gives an energy-unit Hamiltonian. This construction preserves the chosen stochastic clock and does not select physical dynamics or action units. It resolves G04's finite-volume transfer decision.

## 1. Object, domain and transform

Use C124's model:  $N \geq 3$  periodic spins,  $0 \leq b < \infty$ , per-site refresh rate  $a > 0$  and stationary law  $\pi(\sigma) = Z^{-1} \exp(b \sum_i \sigma_i \sigma_{i+1})$ . The generator  $Q = a \sum_i (E_i - I)$  acts on functions in complex  $L^2(\pi)$ ;  $E_i$  is conditional expectation over spin  $i$ . Its time parameter is the auxiliary sampling clock and  $a$  has inverse-time units.

Let  $D = \text{diag } \pi$  and  $U : L^2(\pi) \rightarrow \mathcal{H}_N = (\mathbb{C}^2)^{\otimes N}$  be  $(Uf)_\sigma = \sqrt{\pi(\sigma)} f(\sigma)$ . The codomain has the standard counting inner product;  $U$  is unitary between these two inner-product spaces. All operators below have the entire finite-dimensional space as domain. Set

$$A_N = -UQU^{-1} = -D^{1/2}QD^{-1/2}.$$

Detailed balance makes this matrix Hermitian. Its quadratic form is

$$\langle \psi, A_N \psi \rangle = \frac{1}{2} \sum_{\sigma, \eta} \pi(\sigma) Q_{\sigma\eta} \left| \frac{\psi_\eta}{\sqrt{\pi(\eta)}} - \frac{\psi_\sigma}{\sqrt{\pi(\sigma)}} \right|^2 \geq 0,$$

where diagonal terms vanish. Irreducibility implies a one-dimensional kernel, spanned by the normalized vector

$$|\Omega\rangle = Z^{-1/2} \sum_{\sigma} e^{(b/2) \sum_i \sigma_i \sigma_{i+1}} |\sigma\rangle.$$

Its squared basis amplitudes reproduce the Gibbs probabilities. Regarding those amplitudes and other vectors as physical quantum states requires a state/measurement interpretation in addition to this Hilbert-space map.

## 2. Explicit local terms and their positivity

Write  $Z_i |\sigma\rangle = \sigma_i |\sigma\rangle$  and let  $X_i$  flip spin  $i$ . Put  $\theta = \tanh(2b)$ ,  $s = \text{sech}(2b)$ ,  $c_+ = (1 + s)/2$ ,  $c_- = (s - 1)/2$ . Then

$$A_N = \sum_i h_i, \quad h_i = \frac{a}{2} \left[ I - \frac{\theta}{2} Z_i (Z_{i-1} + Z_{i+1}) - (c_+ I + c_- Z_{i-1} Z_{i+1}) X_i \right].$$

**Derivation.** For fixed neighbours with sum  $S = \sigma_{i-1} + \sigma_{i+1}$ , the original flip rate is  $r_i(\sigma) = a[1 - \sigma_i \tanh(bS)]/2$ . Detailed balance gives transformed off-diagonal entry  $-\sqrt{r_i(\sigma)r_i(\sigma^i)} = -a/[2 \cosh(bS)]$ . The diagonal entry is unchanged. Since  $S = 0, \pm 2$ , the off-diagonal coefficient equals  $c_+ + c_- \sigma_{i-1} \sigma_{i+1}$ , giving the formula. The two neighbours are distinct even for  $N = 3$ ; all indices are cyclic.

Alternatively  $h_i = a(I - U E_i U^{-1})$ . Each  $E_i$  is an orthogonal projection, so  $h_i \geq 0$ ,  $h_i^2 = a h_i$  and  $h_i |\Omega\rangle = 0$ . Thus the sum is frustration-free, although its overlapping terms need not commute. Each term has norm  $a$  and support in the three sites  $i - 1, i, i + 1$ ; the global density transform has not introduced interactions of growing range. For a fixed neighbour block the local ground vector has components proportional to  $e^{bS/2}, e^{-bS/2}$ , which also checks the square-root and sign conventions. At  $b = 0$ ,  $h_i = a(I - X_i)/2$  and the ground state is the product of plus- $X$  vectors. These statements are direct finite-dimensional derivations.

### 3. Exact gap, normalization and limits (C126)

Unitary equivalence and C124 give

$$\text{gap}(A_N) = a[1 - \tanh(2b)].$$

The corresponding nonzero eigenvector is  $UM$  with  $M(\sigma) = \sum_i \sigma_i$ , orthogonal to  $|\Omega\rangle$  by spin-flip symmetry. This is an all-mode spectral statement, inherited from C124 rather than inferred from locality or the ground state alone.

For a supplied constant  $K > 0$  with action units, define  $H_N = KA_N$ . Its ground energy is zero and its excitation gap in energy units is

$$\Delta_N = Ka[1 - \tanh(2b)].$$

For  $K_N \geq K_0 > 0$ ,  $a_N \geq a_0 > 0$ ,  $0 \leq b_N \leq B < \infty$ ,  $\Delta_N \geq K_0 a_0 [1 - \tanh(2B)]$  uniformly in  $N$ . These are bounds on a sequence of finite-volume Hamiltonians. They do not construct an infinite-volume Hilbert representation or establish a continuum gap. Fixed total refresh rate replaces  $a$  by  $a/N$  and closes the gap. Growing  $b_N$  or vanishing  $K_N$  can close it as well when the other parameters stay fixed.

### 4. Which physical premise has been supplied?

The exact semigroup identity is

$$e^{-tH_N/K} = Ue^{tQ}U^{-1}.$$

It identifies the transformed sampling semigroup with imaginary-time evolution of the constructed operator, in the same time units. Defining the unitary  $e^{-itH_N/K}$  is mathematically possible but is an additional choice of real-time evolution; it is not the original stochastic transition law.

| Input                                   | Use                                                  | What remains to identify physically                      |
|-----------------------------------------|------------------------------------------------------|----------------------------------------------------------|
| Positive Gibbs law and detailed balance | Unitary weighted-space transform and Hermiticity     | Physical state and observable interpretation             |
| Single-site heat-bath generator         | Three-site terms and exact inherited gap             | Why this generator governs physical imaginary time       |
| Per-site rate $a$                       | Overall spectral rate scale                          | Conversion from sampling time to physical time           |
| Supplied action constant $K$            | Converts inverse time to energy and sets phase units | Positive universal action normalization                  |
| Uniform $K, a, b$ bounds                | Finite-volume energy-gap lower bound                 | Their physical origin and any further limit construction |

The classical energy function  $-J \sum_i \sigma_i \sigma_{i+1}$  used to prepare  $\pi$  is not the constructed  $H_N$ : the latter has off-diagonal spin flips and coefficients depending on  $b = \beta J$  and  $Ka$ . Knowledge of  $J$  and  $\beta$  fixes neither the refresh clock nor a universal action constant. For the same  $\pi$ , replacing  $Q$  by  $\lambda Q$ ,  $\lambda > 0$ , preserves the ground state and locality while multiplying every excitation gap by  $\lambda$ . Holding  $Q$  fixed and varying  $K$  independently gives the other normalization freedom. Thus the map yields a conditional stochastic-parent Hamiltonian; an independent physical identification is still a premise.

## 5. Evidence and strategic consequence

Written proof and literature status are recorded separately in the ledger. The B71 audit matches the established detailed-balance parent mapping in Henley and Castelnovo et al.; the explicit Ising term is a derived specialization. The written review checks its coefficients and inherited gap. No novelty is asserted.

G04 stops at the explicit finite-volume operator. Additional Ising spectra would not select its clock or action scale. A further operator task requires an independently specified physical dynamics whose imaginary-time generator can be compared with this one. The quantum track's reversible-descent premise is selected next: test whether a specified interacting mechanical orientation flow closes on C125's moment coordinates. STATE owns that bounded decision.